From Defense to Disclosure: Designing Interactive Systems with Multimodal Adaptive Guidance for LLM-Based Mental Health Counseling

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Abstract

Immature psychological defense mechanisms are unconscious self-protective processes that emerge under emotional conflict and stress, often shaping clients' text-based expression through avoidance, rationalization, or suppression. To understand how AI systems can support clients in noticing such latent patterns, we conducted a formative study with six clinical counselors and four clients, deriving design requirements for structured defense recognition and adaptive reflective guidance. We present DefenseMitigator, an interactive system that helps clients notice latent defensive patterns through defense recognition and adaptive multimodal guidance. Rather than challenging or removing defenses, DefenseMitigator positions AI as a transparent reflective partner that supports clients' psychological self-awareness while preserving their autonomy over emotional disclosure.

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1 Introduction

LLM-based mental health support systems increasingly provide clients with accessible and empathetic interaction [1, 11–13, 16, 18]. Prior work has mainly improved AI-side capabilities, including trustworthiness, empathy, and response quality [2, 3, 14]. However, effective mental health support also depends on how clients express, withhold, or reinterpret their own emotional experiences during interaction. Building on this perspective, we focus on a specific but underexplored challenge: how clients' unconscious defense mechanisms shape what they are able or willing to express in text-based support.

Defense mechanisms are unconscious self-protective processes that arise under emotional conflict and stress [5, 7, 10]. Although often adaptive, immature defenses may appear as avoidance, rationalization, or emotional suppression, limiting clients' ability to recognize and communicate underlying emotional experiences [6, 8, 17]. In face-to-face counseling, counselors may infer such defenses from tone, pauses, facial expressions, and other contextual cues. In text-only LLM support, these cues are largely absent, making defenses difficult to identify when they are concealed behind rational, cooperative, or emotionally neutral replies [4, 9, 15]. As a result, systems may respond to the surface meaning of a client's message without helping the client notice the defensive pattern shaping that message.

To address this gap, we present DefenseMitigator, an interactive system that helps clients notice and reflect on latent defensive patterns during text-based mental health support. The system

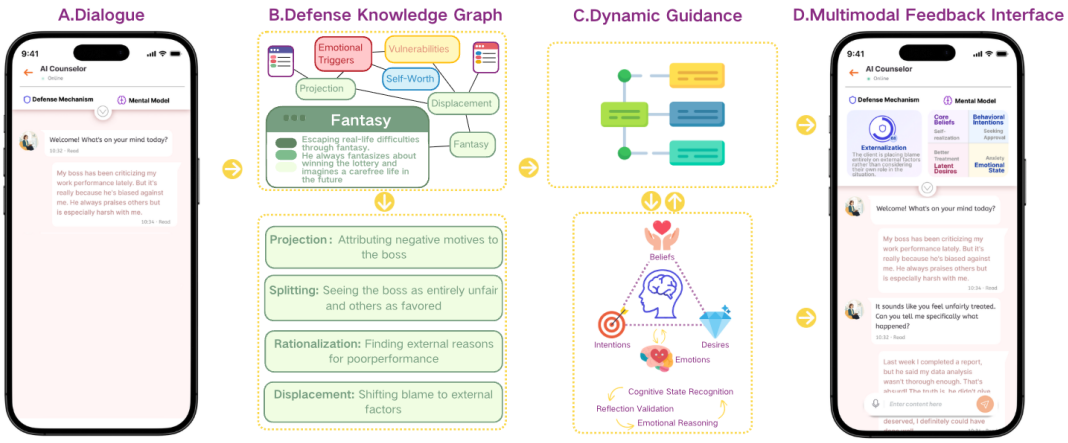


Figure 1: Overview of the DefenseMitigator workflow. (A) A client engages in text-based dialogue with the AI counselor. (B) The dual defense knowledge graph integrates theory-grounded defense concepts with dialogue-derived evidence to identify potential defense mechanisms and provide interpretable cues. (C) The identified defenses inform BDI-based mental-state inference and dynamically adjusted reflective guidance. (D) The system presents adaptive conversational responses and transparent multimodal feedback through the user interface.

combines structured defense recognition with adaptive natural-language and multimodal guidance, enabling clients to examine potentially defensive expressions and gradually explore the emotions beneath them. Rather than confronting clients or attempting to remove defenses, DefenseMitigator provides transparent evidence and progressively adjusted reflective support, allowing clients to regulate the depth and pace of emotional disclosure. Through this work, we highlight psychological self-awareness as a design direction for LLM-based mental health interaction.

2 Formative Study

We conducted semi-structured interviews with 10 participants: six clinical counselors (E1–E6, mean 8 years of experience) and four clients with prior counseling experience (U1–U4). Participants were recruited from 50 social-media respondents and screened for counseling relevance or prior LLM-support experience. Each 40–60 minute session was recorded, transcribed, and thematically analyzed, yielding three design requirements.

DR1: Surfacing text-concealed defenses through structured representation. Counselors noted that text-only interaction removes paralinguistic cues, allowing defenses to appear as rational, cooperative, or emotionally neutral responses (E1, E5), while recognizing such patterns remains "the first and hardest step" (E4). This motivated SD1, Dual-Knowledge-Graph Defense Identification, which links professional defense knowledge with dialogue-derived semantic evidence such as contradictions, hesitation, contextual dependencies, and cross-turn associations.

DR2: Providing defense-sensitive natural-language guidance. Counselors warned that generic prompts or mechanical replies may make clients feel "judged by an algorithm" (E1), potentially increasing defensiveness. This informed the adaptive dialogue component of SD2, Adaptive Multimodal Guidance, which generates reflective prompts and adjusts questioning strategies according to clients' affective and defensive states.

DR3: Supporting progressive disclosure through transparent feedback. Participants emphasized that clients should retain control over the depth and pace of emotional disclosure, and perceived gradual feedback such as emotional fluctuation visualization as helpful for self-paced reflection (U4). This informed the multimodal feedback component of SD2, Adaptive Multimodal Guidance, which combines progressive prompts with transparent visual and optional voice feedback.

3 System Design

The full operational pipeline of DefenseMitigator(Figure 1) adopts a closed-loop two-module architecture, forming a complete cycle from dialogue input (Figure 1-A) to multimodal feedback (Figure 1-D). The first module is the defense knowledge graph identification unit (Figure 1-B), which detects defense types and intensity from textual input. The second module is the interactive guidance and multimodal feedback unit (Figure 1-C), which delivers layered adaptive guidance based on detection results.

SD1: Dual-Knowledge-Graph Defense Identification To address the invisibility of defenses in text-only interaction from raw user dialogue (Figure 1-A), we build a cross-validation framework consisting of a Theory Graph and an Instance Graph, both visualized within the defense knowledge graph component (Figure 1-B). The clinician-validated Theory Graph adopts triple-based formalization, with each defense mechanism as a core entity associated with trigger cause, linguistic manifestation, and internal logic, translating clinical theories into computable structures. The Instance Graph is constructed from 150 hours of annotated counseling dialogues, storing linguistic features and interaction patterns of various defense types.

At runtime, the system extracts three tiers of semantic features for parallel matching: the surface linguistic marker layer, the middle affective tone layer (core), and the deep cognitive logic layer (core). A dynamic weighting algorithm fuses the two matching

scores to output defense judgments with linguistic evidence; as shown in the defense knowledge graph panel (Figure 1-B), defenses such as projection, splitting, and rationalization can be identified simultaneously. Verified new samples are continuously fed back to iteratively improve identification performance.

SD2: Detection-Driven Interactive Guidance Layer This module takes knowledge graph outputs as input, and leverages the BDI reasoning framework in the *Dynamic Guidance* component (Figure 1-C) to model users' latent mental states via Belief–Desire–Intention inference, then forms a self-adaptive closed loop through three intervention tiers.

First, reflective questioning generates tailored open-ended prompts aligned with specific defense types, triggering self-reflection by uncovering three subtle narrative contradictions: disjointed narrative logic, inconsistency between verbal denial and implicit emotional expression, and downplayed negative feelings paired with concurrent somatization. It automatically shifts to empathic reflection when sustained resistance is detected, and the resulting therapeutic dialogue responses are presented to users in the multimodal feedback interface (Figure 1-D).

Second, questionnaire-based progressive prompting adjusts tactics dynamically according to real-time defense intensity: factual-focused closed questions are used in high-defense stages, while scenario-recalling open prompts are adopted in low-defense stages. All inquiries anchor to concrete affective anchors to avoid intellectualization defenses, and the tiered prompt hierarchy is visualized as layered colored blocks within the Dynamic Guidance panel (Figure 1-C).

Third, the visual state dashboard (Figure 1-D) displays two quantitative dimensions via non-judgmental line charts: categorical defense intensity on a 0–10 scale and binary affective valence. This low-threat visual feedback reduces psychological pressure and supports autonomous self-reflection.

4 Discussion and Future work

Our work shifts from optimizing AI understanding to scaffolding users' awareness of self-protective patterns.Future work will integrate longitudinal user modeling to address deeply ingrained relational defense patterns, and develop culturally adaptive intervention frameworks to extend generalizability across diverse populations.

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